PYTHON PROJECT ENVIRONMENT SETUP

Anaconda • Conda • Jupyter Notebook

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THE GOAL

At the end, your Jupyter Notebook must be running from the new Conda environment—not from base.

Before You Start

This guide assumes Anaconda Distribution is already installed on Windows. We will use Anaconda PowerShell Prompt for environment setup and Anaconda Navigator to launch Jupyter Notebook.

ONE PROJECT = ONE ENVIRONMENT

Keep the base environment clean. Your project gets its own environment so its Python version and libraries stay isolated.

Environment name used in this guide:

```
python_project
```

If your instructor gives you a different name, replace python_project everywhere in this guide.

STEP 1 — Open Anaconda PowerShell Prompt

Open the Windows Start Menu and search for Anaconda PowerShell Prompt. Open it.

```
Windows PowerShell
(base) PS C:\Users\Student>
```

CHECKPOINT 1

You should see (base) at the beginning of the prompt. That is normal. We will create a separate environment next.

Now confirm that Conda is available:

```
conda --version
```

Expected: a Conda version number, for example conda 25.x.x or another installed version.

STOP IF THIS FAILS

If you see “conda is not recognized”, stop here. Close the window and make sure you opened Anaconda PowerShell Prompt, not ordinary PowerShell.

STEP 2 — List your Conda environments

Use the command you have already learned in class:

```
conda env list
```

```
(base) PS C:\Users\Student> conda env list
# conda environments:
base C:\Users\Student\anaconda3
```

UNDERSTAND THE *

An asterisk * means that environment is currently active.

Do not worry if you see other environments. We are going to create a new one for this project.

STEP 3 — Create the new project environment

Create a separate environment with Python 3.12:

```
conda create -n python_project python=3.12
```

When Conda asks:

```
Proceed ([y]/n)?
```

Type:

```
y
```

Wait for the installation to finish. Do not close the PowerShell window.

WHAT DID WE JUST DO?

conda create creates an environment. -n python_project gives it a name. python=3.12 tells Conda which Python version to install.

STEP 4 — Verify the environment was created

Run:

```
conda env list
```

```
(base) PS C:\Users\Student> conda env list
# conda environments:
base C:\Users\Student\anaconda3
python_project C:\Users\Student\anaconda3\envs\python_project
```

IMPORTANT

The environment appearing in this list means it exists. It does not yet mean we are working inside it.

STEP 5 — Activate the project environment

```
conda activate python_project
```

Look carefully at the PowerShell prompt.

```
BEFORE:
(base) PS C:\Users\Student>

AFTER:
(python_project) PS C:\Users\Student>
```

CHECKPOINT 2

The prompt must now begin with (python_project). If it still says (base), do not continue.

STEP 6 — Prove that the correct environment is active

First run:

```
conda env list
```

```
(python_project) PS C:\Users\Student> conda env list
# conda environments:
base C:\Users\Student\anaconda3
python_project * C:\Users\Student\anaconda3\envs\python_project
```

THE * MUST BE HERE

The asterisk must be beside python_project.

Now verify the Python version:

```
python --version
```

Expected: Python 3.12.x.

Finally, find the Python executable being used:

```
where.exe python
```

The first path should point into:

```
...\anaconda3\envs\python_project\python.exe
```

STEP 7 — Install Jupyter Notebook + ipykernel

We need Jupyter Notebook and the Python kernel package inside the project environment.

```
conda install notebook ipykernel
```

When asked to proceed, type:

```
y
```

Wait until Conda reports that the transaction has completed.

WHY ipykernel?

Jupyter needs a kernel to execute Python code. The kernel package lets our python_project environment appear as a selectable Python environment inside Jupyter.

STEP 8 — Verify Jupyter installation

Run each command separately:

```
jupyter --version
```

```
jupyter notebook --version
```

```
python -m ipykernel --version
```

CHECKPOINT 3

Each command should return version information instead of a “not recognized” error.

STEP 9 — Register the environment as a Jupyter kernel

Run this while (python_project) is still active:

```
python -m ipykernel install --user --name python_project --display-name "Python
(python_project)"
```

Then verify the kernel list:

```
jupyter kernelspec list
```

WHAT WE CREATED

We created a Jupyter option called Python (python_project). This is what we will select when creating the project notebook.

STEP 10 — Open Jupyter through Anaconda Navigator

From this point onward, use Anaconda Navigator to open Jupyter Notebook. Do not launch Jupyter from the PowerShell prompt for the student workflow.

1. Open Anaconda Navigator
2. Find the Jupyter Notebook application
3. Click Launch
4. Wait for Jupyter to open in your browser

NOTE ABOUT VERSIONS

Anaconda's current releases may show Jupyter Notebook/JupyterLab differently in Navigator. The exact tile can vary by installed version. The goal here is to launch the Jupyter Notebook application through Navigator.

STEP 11 — Create a new .ipynb notebook

In the Jupyter interface, use the Add/New option to create a new notebook.

Choose a Python notebook and, when Jupyter asks for the kernel/environment, select:

```
Python (python_project)
```

DO NOT SKIP THIS

Creating the notebook is not enough. You must make sure the notebook is attached to Python (python_project).

STEP 12 — FINAL VERIFICATION

This is the most important test in the entire guide. In the first notebook cell, type:

```
import sys

print(sys.executable)
print(sys.version)
```

Run the cell.

```
Notebook output:
C:\Users\Student\anaconda3\envs\python_project\python.exe
3.12.x ...
```

GREEN LIGHT — READY FOR THE PROJECT

If the executable path contains envs\python_project\python.exe, your notebook is running from the correct Conda environment.

Troubleshooting — Stop, Identify, Fix

conda is not recognized	You opened ordinary PowerShell. Open Anaconda PowerShell Prompt.
Still seeing (base)	Run: conda activate python_project
No * beside python_project	Run: conda activate python_project, then conda env list
Wrong Python path	Run where.exe python and check the first path.
jupyter is not recognized	Make sure python_project is active, then install notebook + ipykernel.
Python (python_project) is missing	Run the ipykernel registration command again.
Notebook opens but runs another Python	Select Python (python_project), then rerun the sys.executable check.

THE GOLDEN TEST

Inside the notebook, print(sys.executable) must point to your project environment. This is more reliable than simply looking at a menu label.

One-Page Command Cheat Sheet

Use this only after you understand the step-by-step setup.

```
conda --version
conda env list

conda create -n python_project python=3.12
conda activate python_project

conda env list
python --version
where.exe python

conda install notebook ipykernel

jupyter --version
jupyter notebook --version
python -m ipykernel --version

python -m ipykernel install --user --name python_project --display-name "Python
(python_project)"
jupyter kernelspec list
```

Then:

Anaconda Navigator → Jupyter Notebook → Add/New → New .ipynb → Python (python_project)

Final notebook test:

```
import sys
print(sys.executable)
print(sys.version)
```

SUCCESS CONDITION

The printed path contains envs\python_project\python.exe.

Sources & Notes

The command sequence and environment concepts were checked against current official Conda and Anaconda documentation. The interface can vary slightly with Anaconda Navigator/Jupyter versions, so the guide deliberately uses the underlying verification commands as the final authority.

Official references used:

• Conda documentation — environment management and Python environments.

• Anaconda Navigator — current Navigator capabilities and application launching.

• Anaconda — beginner environment guide and Jupyter launch workflow.

IMPORTANT

This guide intentionally uses conda env list throughout, matching the terminology used in your classroom. It does not teach conda info --envs.